

6.7 Determining Efficiencies

Efficiency is how well we detect particles as the correct ones (e.g. electrons).

Tag and probe: strict selection criteria on the tagged electron, the second one becomes the probe, which is used to determine efficiency.

- tag and probe method
- tag has to pass selection criteria
- probe has to be checked (is it an electron?)
- efficiency is fraction of probes passing

$$\epsilon_{total} = \epsilon_{reconstruction} \times \epsilon_{identification} \times \epsilon_{trigger} \times \epsilon_{additional}$$